HYBRID RADIUS MANAGEMENT SYSTEM

Centralized Admin Control with Driver-Level Custom Radius Flexibility

ENHANCEMENT OF THE EXISTING SYSTEM
Admin stays the authority. Drivers get controlled flexibility.

1 ADMIN PANEL

Centralized configuration

Default Radius — EXISTING
5 KM

Minimum Radius — NEW
2 KM

Maximum Radius — NEW
15 KM

Enable Driver Custom Radius — NEW
ON / OFF

Admin keeps complete control: universal default, allowed range and the ON/OFF switch.

2 DRIVER APP

Service Radius Settings

Selected radius
8 KM

Save Radius

2 KM min — 5 default — 15 KM max

Driver workflow
1 Open Driver App > Settings > Service Radius
2 See Admin min / max limits
3 Choose a preferred radius, then Save
4 Backend validates and stores the choice

Custom Radius OFF: control hidden or disabled, Admin default used.

Config sync

Driver preference

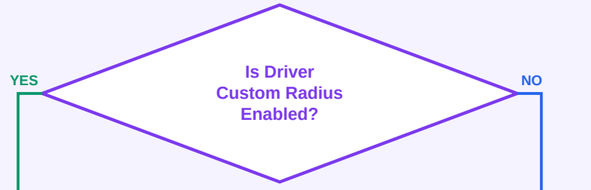
3 BACKEND

Radius resolution and validation

Configuration store (existing schema)
Admin settings: default, min, max, allowCustom
Driver profile: serviceRadius, isCustomRadius
Reuses existing records, no new collections

RADIUS RESOLUTION
1 Fetch Admin config
2 Check driver custom preference

```
allowCustomRadius  AND  driver.isCustomRadius
```



YES
Use driver custom radius, kept within Admin min and max

NO
Use Admin default radius (custom value stays stored but inactive)

EFFECTIVE RADIUS

VALIDATION (server side)
- Radius below Admin minimum: rejected
- Radius above Admin maximum: rejected

Frontend checks never replace this

Effective radius

4 RIDE MATCHING ENGINE

Existing engine, effective radius applied

Receives the effective radius — ENHANCED
One resolved value for each driver

Finds eligible ride requests — ENHANCED
Pickup distance is within the driver's effective radius

Existing eligibility rules — UNCHANGED
Zone, vehicle type, wallet, online status and the other current checks still apply

Booking, dispatch and assignment — UNCHANGED
Existing workflows are not modified

Eligible requests

5 FINAL RESULT

Driver-level radius-based matching

Illustrative example

Driver A	3 KM — Driver custom radius
Driver B	8 KM — Driver custom radius
Driver C	5 KM — Admin default
Driver D	10 KM — Driver custom radius

Each driver receives only the requests inside their effective radius.

ELIGIBLE RIDE REQUESTS
Matching follows driver-level radius, with Admin limits and the Admin default always enforced.

How the existing system is extended

KEPT AS IS
Admin's centralized default radius, the Admin Panel and the whole ride-matching, dispatch and assignment flow.

ADDED
Admin min / max / ON-OFF switch, a driver service radius preference, and one backend resolution step.

RESULT
Existing behavior is the fallback. When Custom Radius is OFF, or a driver never chooses, the Admin default applies.

Field names in this document are illustrative and should be mapped to the existing schema (for example the existing Admin driver search radius setting and the driver profile).

BUSINESS RULES, DATA STRUCTURES AND BENEFITS

Hybrid Radius Management System: how the rules behave and what changes in the data

IMPORTANT BUSINESS SCENARIOS

1 New driver registration

- Admin default radius is assigned automatically.
- isCustomRadius stays false.

2 Driver selects a custom radius

- Preference is saved after backend validation.
- It is applied during matching.

3 Admin changes the default radius

- Drivers on the default get the new value.
- Active custom preferences are not overwritten.

4 Admin disables Custom Radius

- All drivers use the Admin default at once.
- Custom preferences stay stored but inactive.

5 Admin re-enables Custom Radius

- Saved preferences become active again.
- Re-validated against the current min and max.

6 Invalid radius selection

- Values outside the permitted range are rejected by the backend.
- No change is stored.

EFFECTIVE RADIUS BY STATE

Admin switch	Driver preference	Effective radius
Custom Radius ON	Custom radius saved	**Driver radius (within Admin min and max)**
Custom Radius ON	No custom radius	**Admin default radius**
Custom Radius OFF	Custom radius saved	**Admin default (preference kept, inactive)**
Custom Radius OFF	No custom radius	**Admin default radius**

DATA ARCHITECTURE

Admin configuration (illustrative)

```
{
  defaultRadius: 5,
  minRadius: 2,
  maxRadius: 15,
  allowCustomRadius: true
}
```

Driver configuration (illustrative)

```
{
  serviceRadius: 8,
  isCustomRadius: true
}
```

Resolution rule

```
IF allowCustomRadius
   AND driver.isCustomRadius:
  effective = driver.serviceRadius
              (within min..max)
ELSE:
  effective = admin.defaultRadius
```

Mapping note

These are illustrative structures. Map them onto the existing Admin settings and the existing driver record instead of creating new collections or tables. The Admin default can reuse the current Admin driver search radius setting.

KEY BENEFITS

- ✓ Existing centralized Admin system remains intact
- ✓ Drivers get independent radius flexibility
- ✓ Admin keeps control of minimum and maximum limits
- ✓ Admin can enable or disable custom radius at any time
- ✓ Existing custom preferences can be preserved
- ✓ Minimal backend and database changes
- ✓ No need to replace the existing ride-matching architecture